

2024 HKDSE Physics

Paper 1B — Structured Questions

Original questions and official answers with detailed English reasoning. Calculations show relationships, substitution and units; each structured subpart is presented separately.

Editable questions with original diagrams, official answers and detailed English reasoning. Each subpart is separated for clarity.

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Take $g = 9.81 \text{ m s}^{-2}$.

Official answers are placed directly below each question.

Question 1

A bottle of water of temperature 4°C is taken out from a refrigerator and placed on a table. The room temperature is 30°C .

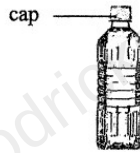


Figure 1.1

- (a) State one way of heat transfer from the surroundings to the bottle of water. (1 mark)
- (b) Explain whether wrapping the bottle with an aluminum foil will speed up the rate of temperature increase. (2 marks)
- (c) After a period of time, water droplets form on the surface of the bottle.
 - (i) State the source of the water droplets. (1 mark)
 - (ii) Suppose 0.40 g of water droplets are collected on the surface of the bottle. Estimate the latent heat released from these droplets. (2 marks)

Given: specific latent heat of vaporization of water $= 2.26 \times 10^6 \text{ J kg}^{-1}$

Official marking scheme

Paper 1 Section B

Solution	Marks	Remarks
1. (a) conduction / radiation	1A	
	1	
(b) Will not, as the silvery surface of aluminum foil is a poor radiation absorber / good radiation reflector.	1A+1A	
	2	
(c) (i) The water droplets come from the moisture / water vapour in the air (surroundings).	1A	
	1	
(ii) $E = (0.40 \times 10^{-3})(2.26 \times 10^6)$ $= 904 \text{ J}$	1M 1A	
	2	

(a) Radiation from the warmer surroundings to the colder bottle is one accepted mode. Conduction through contact with warmer material is also possible.

(b) No. The shiny aluminium surface is a poor absorber and a good reflector of thermal radiation, so it reduces radiative energy gain from the surroundings. It does not speed up the warming by this mechanism.

(c)(i) Water vapour in the surrounding air condenses on the cold bottle; the droplets do not pass through the bottle wall.

(c)(ii) Convert 0.40 g to $0.40 \times 10^{-3} \text{ kg}$. The latent heat released is $Q = ml = (0.40 \times 10^{-3})(2.26 \times 10^6) = 904 \text{ J}$.

Question 2

*Figure 2.1 shows a closed vessel of fixed volume containing argon, a monatomic gas, maintained at temperature T_0 . The root-mean-square speed (c_{rms}) of the argon atoms inside is 500 m s^{-1} .

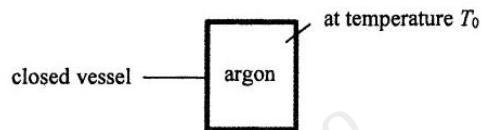


Figure 2.1

- (a) (i) Calculate the average kinetic energy of the argon atoms, E_k . Given that the mass of an argon atom is $6.63 \times 10^{-26} \text{ kg}$. (2 marks)
- (ii) Hence, or otherwise, find T_0 (in K). (2 marks)
- (b) If some of the argon has been leaked from the vessel while the temperature remains at T_0 , state and explain the change, if any, in the root-mean-square speed of the argon atoms inside the vessel. (2 marks)

Official marking scheme

2. (a) (i) $E_k = \frac{1}{2} (6.63 \times 10^{-26}) (500)^2$ $= 8.2875 \times 10^{-21} \text{ J} \approx 8.29 \times 10^{-21} \text{ J}$	1M 1A 2	Accept: $(8.29 \sim 8.30) \times 10^{-21} \text{ J}$
(ii) $E_k = \frac{3RT_0}{2N_A}$ $T_0 = \frac{2}{3} \left(\frac{8.29 \times 10^{-21}}{8.31} \right) (6.02 \times 10^{23})$ $= 400.247 \text{ (K)} \approx 400 \text{ (K)}$	1M 1A 2	
(b) As temperature remains at T_0 , E_k remains unchanged / $c_{\text{r.m.s.}}$ depends on temperature, so root-mean-square speed $c_{\text{r.m.s.}}$ remains unchanged.	1A 1A 2	e.c.f. from (a)(i)

(a)(i) The mean translational kinetic energy is $E_k = \frac{1}{2} m c_{\text{rms}}^2 = \frac{1}{2} (6.63 \times 10^{-26}) (500)^2 = 8.2875 \times 10^{-21} \text{ J} \approx 8.29 \times 10^{-21} \text{ J}$.

(a)(ii) For monatomic gas, $E_k = 3RT_0/(2N_A)$. Thus $T_0 = 2N_A E_k / (3R) = 2(6.02 \times 10^{23})(8.2875 \times 10^{-21}) / (3 \times 8.31) \approx 400 \text{ K}$.

(b) The root-mean-square speed is unchanged. Since $c_{\text{rms}} = \sqrt{(3RT_0/M)}$, it depends on temperature and molar mass, not on the number of atoms remaining. The total gas energy can decrease while the average kinetic energy per atom stays the same.

Question 3

A small block P of mass 0.20 kg rests on side A of a horizontal table as shown. The surface of the table is smooth from A to B but is rough from B to C. Neglect air resistance.

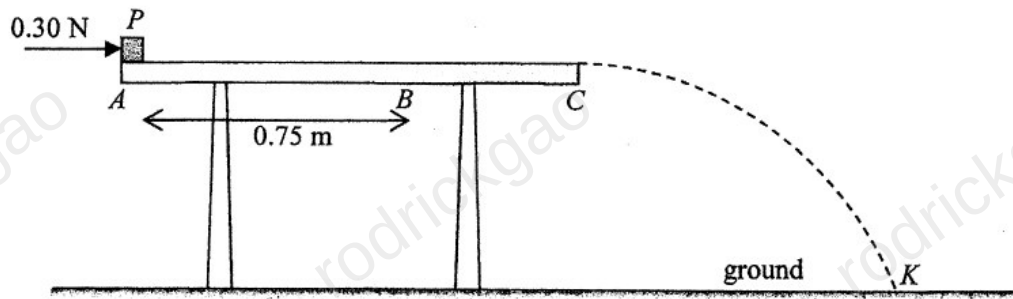


Figure 3.1

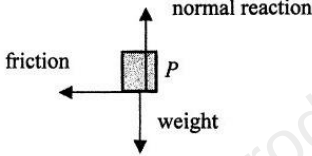
A horizontal force of 0.30 N acts on the block as shown.

- Find the speed of the block after it travels 0.75 m to point B. (2 marks)
- The horizontal force is removed when block P just reaches B. When the block continues to travel from B to C, on the free-body diagram below indicate and name the force(s) acting on the block. (2 marks)



- *(c) The block leaves side C of the table and takes 0.35 s to reach point K on the ground. Estimate the height of the table. (2 marks)

Official marking scheme

Solution	Marks	Remarks
3. (a) By $Fs = \frac{1}{2}mv^2$, $0.30 \times 0.75 = \frac{1}{2}(0.20)v^2$ $v = 1.5 \text{ m s}^{-1}$	1M 1A 2	Or $v^2 = u^2 + 2as$ and $F = ma$ $v^2 = 0^2 + 2\left(\frac{0.30}{0.20}\right)(0.75)$
(b) 	2A 2	Accept: normal force, gravity
(c) Height = $\frac{1}{2}gt^2 = \frac{1}{2}(9.81)(0.35)^2$ $= 0.6008625 \text{ m} \approx 0.601 \text{ m}$	1M 1A 2	Accept: 0.60 m ~ 0.613 m

(a) Between A and B, the surface is smooth and the horizontal force does work on the initially stationary block. $Fs = \frac{1}{2}mv^2$ gives $0.30 \times 0.75 = \frac{1}{2}(0.20)v^2$, hence $v = 1.50 \text{ m s}^{-1}$.

(b) Draw three arrows from P: weight vertically down, normal reaction vertically up, and friction horizontally left, opposite its motion. The applied 0.30 N force has already been removed. Vertical acceleration is zero, so normal reaction equals weight.

(c) The block leaves the table horizontally, so its initial vertical velocity is zero. Its height is $h = \frac{1}{2}gt^2 = \frac{1}{2}(9.81)(0.35)^2 = 0.6009 \text{ m} \approx 0.60 \text{ m}$. Horizontal slowing before C does not change this vertical calculation.

Question 4

*A sphere of radius 0.50 m rotates about a vertical axis ACB with a constant angular speed of 6 rad s^{-1} , with C being the sphere's centre. A small plasticine P of mass 0.020 kg is stuck on the surface of the sphere such that it rotates on a horizontal plane containing C as shown in Figure 4.1(a).

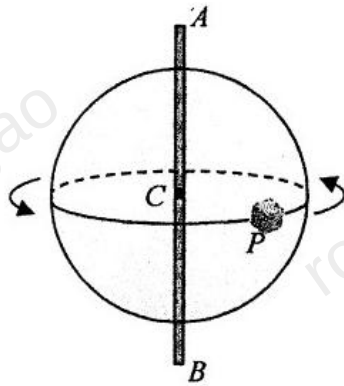


Figure 4.1(a)

Top view from A

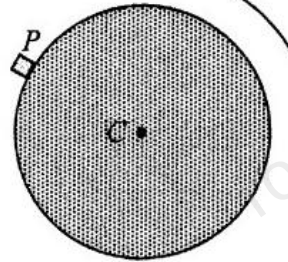


Figure 4.1(b)

- (a) (i) Find the magnitude of the centripetal force acting on P. (2 marks)
- (ii) The sphere's angular speed is increasing. If P finally leaves the sphere at the position shown in Figure 4.1(b) (top view from A), indicate on Figure 4.1(b) the direction of motion of P when it just leaves the sphere. (1 mark)
- (b) Another plasticine Q is stuck on the sphere's surface as shown in Figure 4.2.

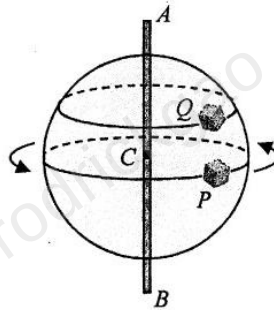
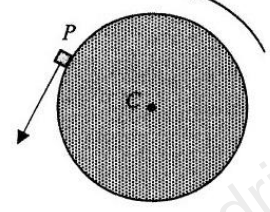


Figure 4.2

- (i) State whether P and Q have the same angular speed. (1 mark)
- (ii) Explain whether P and Q have the same magnitude of centripetal acceleration. (2 marks)

Official marking scheme

4. (a) (i) $F = m\omega^2 r = 0.020 \times 6^2 \times 0.50 = 0.36 \text{ N}$	1M 1A 2	
(ii) Top view from A 	1A 1	
(b) (i) same angular speed	1A 1	
(ii) different / smaller magnitude of centripetal acceleration ($a_Q < a_P$) as Q takes a different / smaller radius ($r_Q < r_P$)	1A 1A 2	

(a)(i) The circular-path radius is 0.50 m. $F = m\omega^2 r = 0.020 \times 6^2 \times 0.50 = 0.36 \text{ N}$, directed towards the rotation axis.

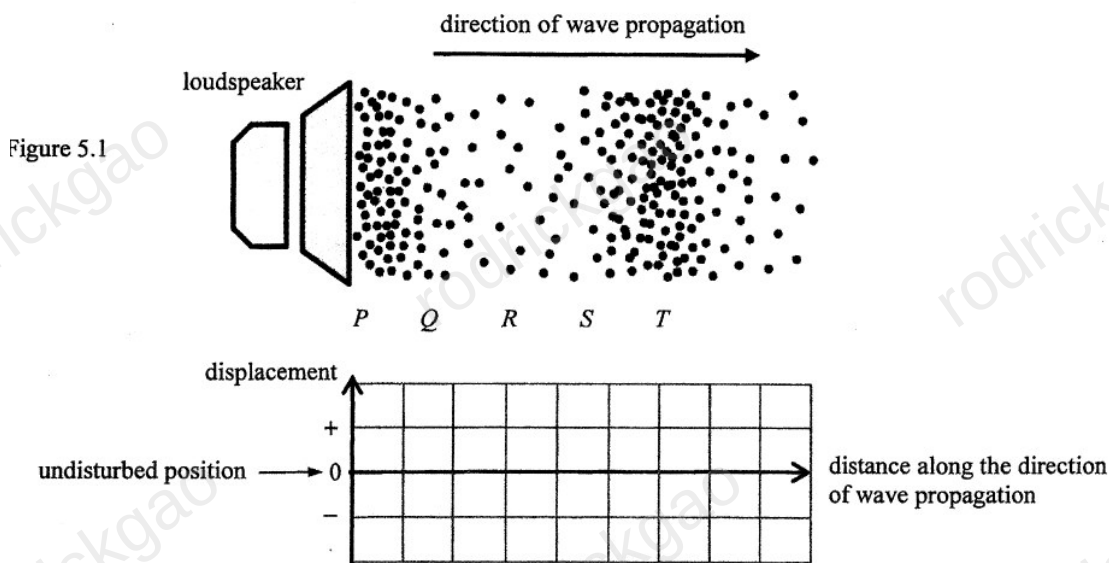
(a)(ii) Draw the velocity tangent to the circle at P in the indicated anticlockwise sense: downwards and leftwards in the top-view diagram. It is not radially outward.

(b)(i) P and Q have the same angular speed because they rotate together on the rigid sphere.

(b)(ii) Their centripetal accelerations differ. Q is closer to the vertical rotation axis, so $r_Q < r_P$. Since $a = \omega^2 r$ at the same ω , $a_Q < a_P$. Use distance to the axis, not distance to the sphere's centre.

Question 5

Figure 5.1 represents the distribution of air particles at a certain instant in a sound wave produced by a loudspeaker.



- (a) Sketch the displacement-distance graph of the air particles at that instant in the grid provided. Take the displacement to the right as positive (+). (2 marks)
- (b) At the instant shown, state a region (P, Q, R, S or T) in which
- the air particles are moving with the highest speed; (1 mark)
 - the air pressure is the lowest. (1 mark)

Official marking scheme

Solution	Marks	Remarks
5. (a) displacement from undisturbed position distance along the direction of wave propagation	2A	
	2	
(b) (i) P / R / T	1A	
	1	
(ii) R	1A	
	1	

(a) Draw a smooth sinusoidal displacement curve crossing zero at P with negative slope, reaching a negative maximum at Q, crossing zero upward at R, reaching a positive maximum at S, and crossing zero downward at T. A compression occurs where displacement decreases with position; a rarefaction occurs where it increases.

(b)(i) P, R or T is acceptable. Particles at these positions are passing through equilibrium and have their greatest instantaneous speed.

(b)(ii) R is the centre of a rarefaction, where the particle density and air pressure are lowest.

Question 6

*After removing the protective coating and the reflective layer of a compact disc (CD), the CD becomes a diffraction grating with evenly spaced parallel slits (Figure 6.1(a)). The set-up in Figure 6.1(b) employs a laser beam which can be used to measure the slit separation d .

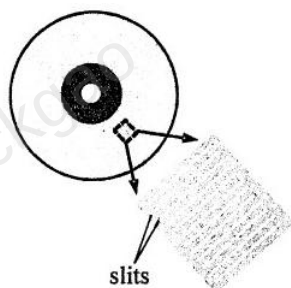


Figure 6.1(a)

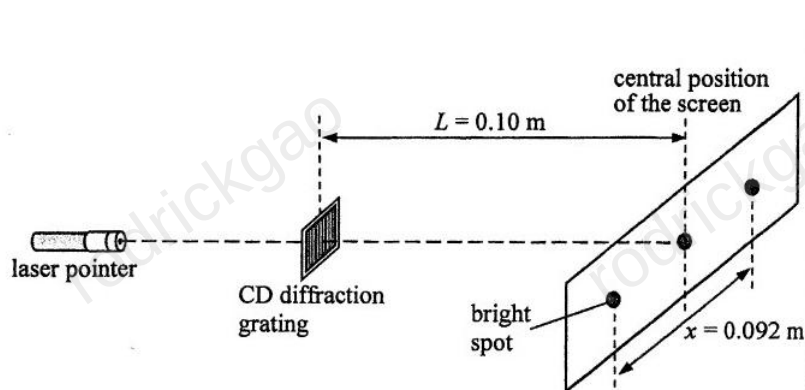


Figure 6.1(b)

Given that the separation between the CD diffraction grating and the screen, L , is 0.10 m, the separation between the first-order maxima, x , is 0.092 m and the wavelength λ of the red laser light beam used is 650 nm.

(a) Estimate d . (3 marks)

(b) State and explain the change in x if green laser light is used. (2 marks)

Official marking scheme

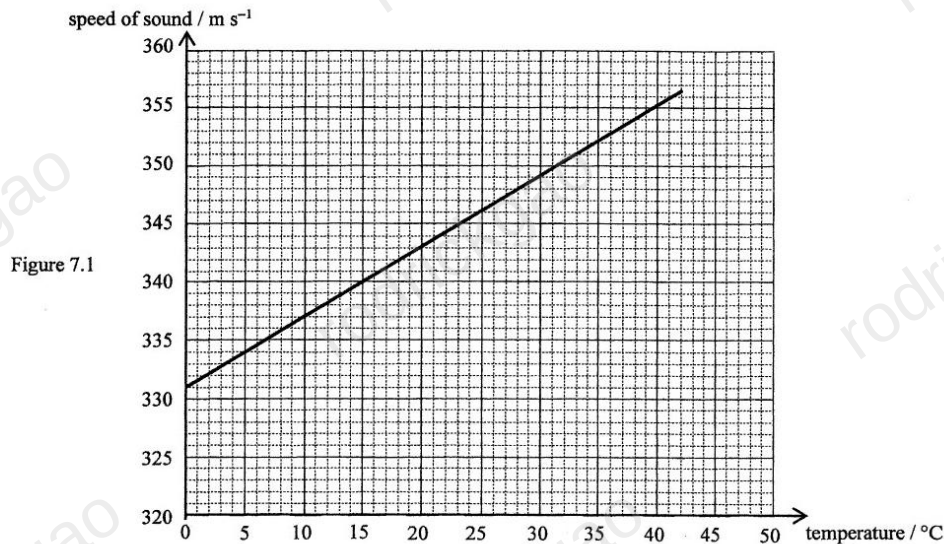
6. (a) Apply $d \sin \theta = m\lambda$, where θ is the deviation angle of the diffracted beam, m is the order of diffraction maximum. $\tan \theta = \frac{x/2}{L} \Rightarrow \theta = 24.702430^\circ$ or $\sin \theta = \frac{x/2}{\sqrt{L^2 + (x/2)^2}} = 0.417906$ $d = \frac{\lambda}{\sin \theta} = \frac{650 \times 10^{-9}}{0.418}$ $= 1.555375 \times 10^{-6} \text{ m} \approx 1.56 \mu\text{m}$ (b) x will decrease as λ decreases, $\sin \theta$ and θ will decrease.	1M 1M 1A 3 1A 1A 2	Accept: $1.56 \mu\text{m} \sim 1.6 \mu\text{m}$
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(a) The distance from the central maximum to either first-order maximum is $x/2 = 0.046$ m. Thus $\tan \theta = 0.046/0.10$ and $\theta = 24.70^\circ$. For first order, $d \sin \theta = \lambda$; hence $d = (650 \times 10^{-9})/\sin 24.70^\circ = 1.56 \times 10^{-6} \text{ m} = 1.56 \mu\text{m}$. The angle is too large to replace $\sin \theta$ by $\tan \theta$ for an accurate result.

(b) x decreases. Green light has a shorter wavelength; with d unchanged, $\sin \theta = \lambda/d$ decreases. The smaller diffraction angle gives a smaller separation $x = 2L \tan \theta$.

Question 7

The graph below shows the speed of sound at different air temperatures.



- (a) With reference to the graph, state the relationship between the speed of sound and the air temperature. (1 mark)
- (b) Figure 7.2 shows sound travelling from a distant tower clock to an observer at night when the ground cools.

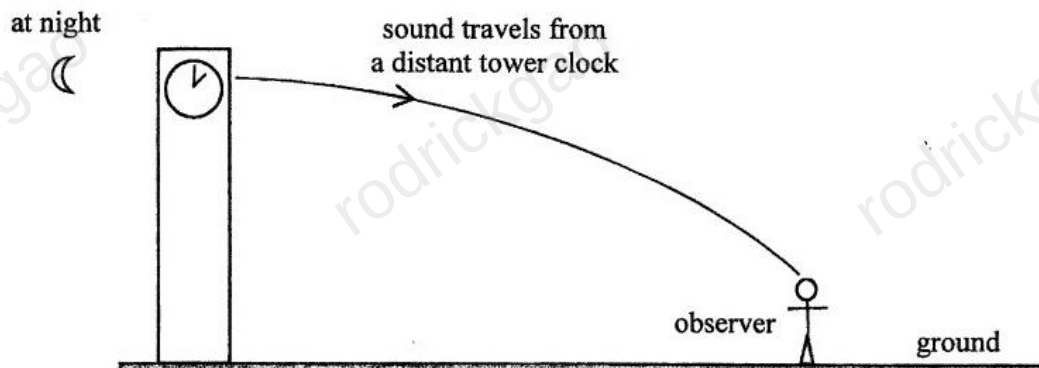


Figure 7.2

How does the temperature difference between air layers above the ground explain the situation in Figure 7.2? (2 marks)

- (c) It is known that the air temperature is 10 °C. Referring to the graph in Figure 7.1:
- estimate the ratio of speed of light to the speed of sound; (1 mark)
 - estimate the distance of a thunderstorm from an observer O if thunder is heard 3.0 s after lightning is seen by O. (2 marks)

Official marking scheme

Solution	Marks	Remarks
7. (a) The speed of sound increases (linearly) with air temperature. Or The higher the air temperature, the greater is the speed of sound.	1A 1	
(b) The air near the ground is cooler while the air layers higher above are relatively warmer, the sound wave bends downward (towards the normal) as it travels slower when going from a higher temperature (upper) layer into a low temperature (lower) layer and vice versa, i.e. refraction occurs.	1A 1A 2	
(c) (i) Speed ratio $\frac{3.00 \times 10^8}{337} = 8.90208 \times 10^5 \approx 8.90 \times 10^5$	1A 1	
(ii) The distance is given by 337×3.0 $= 1011 \text{ m}$	1M 1A 2	e.c.f. from (c)(i) using incorrect sound speed Accept: 1010 m ~ 1011 m

(a) The speed of sound increases approximately linearly with air temperature over the range shown. It is not directly proportional to Celsius temperature because the line does not pass through the origin.

(b) At night the air near the ground is cooler and sound travels more slowly there than in the warmer upper layers. Refraction therefore curves the sound path downwards towards the slower region, allowing sound to reach the observer as drawn.

(c)(i) Read $v \approx 337 \text{ m s}^{-1}$ at 10°C . The ratio is $c/v = (3.00 \times 10^8)/337 = 8.90 \times 10^5$.

(c)(ii) Light travel time is negligible on this scale. The distance is approximately $vt = 337 \times 3.0 = 1011 \text{ m} \approx 1.01 \text{ km}$. More exactly, $3.0 = d/v - d/c$, which gives practically the same result.

Question 8

Figure 8.1 shows a pencil in a glass of water with an observer viewing vertically downwards. Given that the refractive index of water is 1.33.

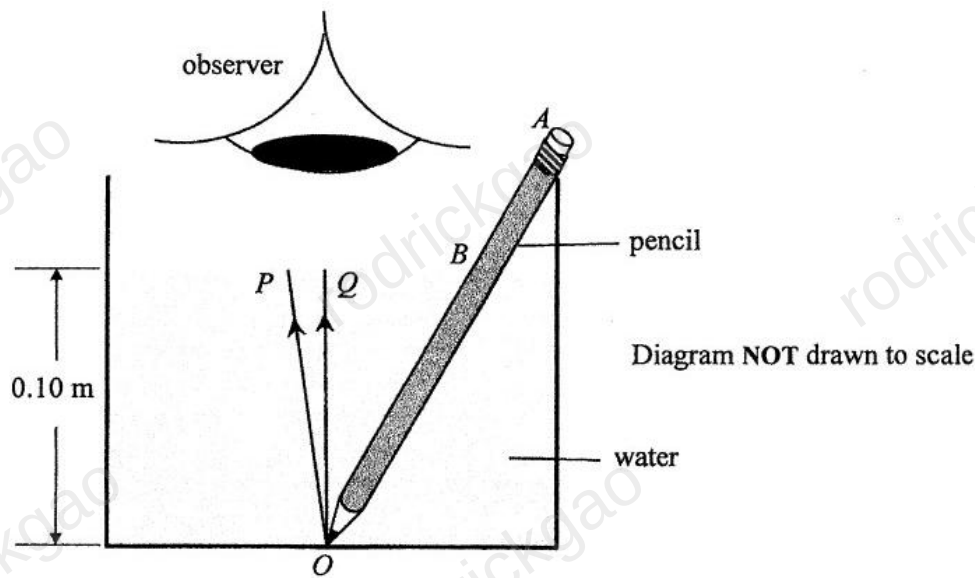
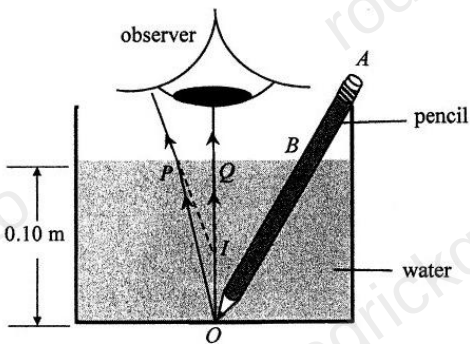


Figure 8.1

Light rays P and Q come from the tip O of the pencil, with Q along the vertical.

- On Figure 8.1, complete the paths of these light rays and indicate the image of O seen by the observer (label this as I). (2 marks)
- The angle between ray P and ray Q is 1.5° . Find the angle of refraction of ray P. (2 marks)
- Hence, or otherwise, find the distance between image I and the water surface if the depth of water is 0.10 m. (2 marks)

Official marking scheme

8. (a)  (b) $1.33 = \frac{\sin r}{\sin 1.5^\circ}$ $r = 1.995175^\circ \approx 2.00^\circ$ (c) $\frac{h_1}{h} = \frac{\tan 1.5^\circ}{\tan 2.00^\circ} = 0.751681 \text{ (or } \tan 1.5^\circ = \frac{PQ}{0.10} \text{ \& } \tan 2^\circ = \frac{PQ}{h_1} \text{)}$ $\Rightarrow h_1 \approx 0.075 \text{ m (i.e. 7.5 cm)}$	1A 1M 2 1M 1A 2 1M 1A 2	1M for image <i>I</i> formed by the two refracted rays. e.c.f. from (b), explicit step / method shown Accept: $\frac{\text{real depth}}{\text{apparent depth}} = \frac{h}{h_1} = n$
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(a) Ray Q is normal to the surface and continues straight. Ray P bends away from the normal on leaving water. Extend the two emergent rays backwards with dashed lines; their intersection above O and below the surface is the virtual image I.

(b) Snell's law gives $1.33 \sin 1.5^\circ = \sin r$. Therefore $r = 1.995^\circ \approx 2.00^\circ$. The angle is measured from the normal, not from the surface.

(c) For near-normal viewing, refractive index = real depth/apparent depth. Thus the image depth is $0.10/1.33 = 0.0752 \text{ m} \approx 7.5 \text{ cm}$. Equivalently, $h = 0.10 \tan 1.5^\circ / \tan r$ gives about 0.075 m.

Question 9

Figure 9.1 shows two identical conducting light spheres P and Q placed close to each other without touching.

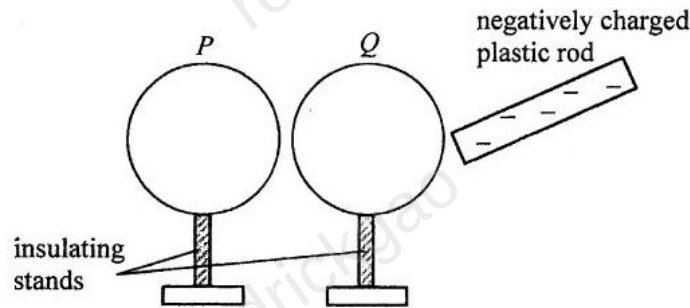


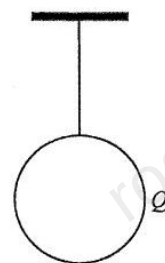
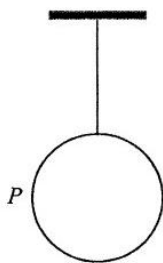
Figure 9.1

Both spheres are uncharged initially. A negatively charged plastic rod is brought close to Q but without touching the sphere.

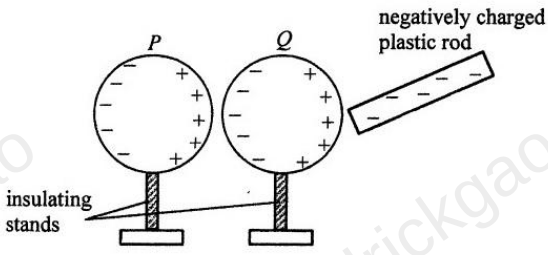
- (a) On Figure 9.1, sketch the distribution of charges on P and on Q. (2 marks)

A student touches sphere P momentarily with his finger while the plastic rod is in the position as shown. The plastic rod is removed afterwards.

- (b) Explain whether the amount of charges on the plastic rod would decrease. (2 marks)
- (c) The two light spheres are now suspended by nylon threads from fixed supports such that the spheres are far apart from each other. Explain how you would use one of the spheres to test for the unknown polarity of another charged rod. (2 marks)



Official marking scheme

Solution	Marks	Remarks
9. (a) 	2A	
	2	
(b) No, it remains unchanged as charging is through induction (without any contact or transfer of charges).	1A+1A	
	2	
(c) Hold / bring the unknown charged rod (assume positive) near sphere P, which carries positive charges, without touching it.	1A	
If P is repelled by the rod, the rod should carry positive charges.	1M	1M for the correct procedure to identify the polarity of the charged rod.
<u>And / Or</u> If P is attracted by the rod, the rod should carry negative charges.		
	2	

(a) On each sphere, place negative charges on the left and positive charges on the right. The negative rod repels mobile electrons away from it; each isolated sphere is polarized but remains neutral overall before earthing. Show equal amounts of positive and negative charge on each sphere.

(b) The rod's charge is unchanged. It never touches either sphere, so no charge is transferred to or from it. During earthing, electrons can leave P through the finger to Earth; this does not remove electrons from the rod.

(c) Use P, which retains a positive net charge after the finger is removed and then the inducing rod is taken away. Bring the unknown charged rod near P without contact. Repulsion identifies a positive rod; attraction indicates a negative rod in the intended test with a known charged sphere and a rod known to be charged. Q remains neutral and is not a suitable polarity reference.

Question 10

- (a) As shown in Figure 10.1, one end of an inextensible string is attached to a small metal ball while its upper end O is fixed to a stand. A protractor is fixed onto the stand behind the string.

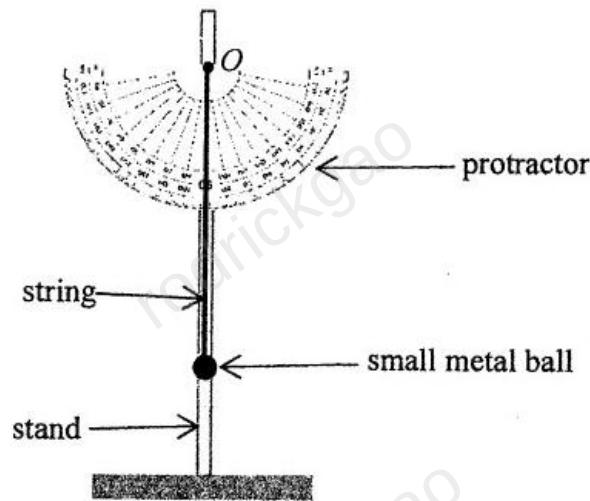


Figure 10.1

Write down the steps that you would take to demonstrate the conservation of energy using the set-up given. (4 marks)

- (b) Figure 10.2 shows a Newton's cradle which consists of five identical metal spheres suspended in a row by inextensible light strings of equal length from a fixed horizontal support.

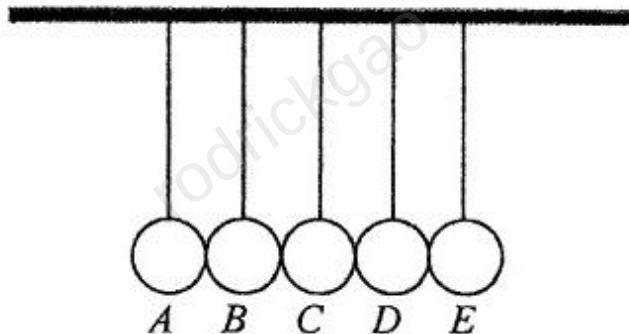


Figure 10.2

Sphere A and B together are displaced slightly to the left and then released from rest. Just after collision, both A and B come to rest while sphere D and E swing together to the right slightly. Neglect air resistance.

- (b)(i)(I) Even though the tensions in the strings acting on the spheres are external forces, explain why the law of conservation of momentum can still be applied for this collision. (1 mark)
- (b)(i)(II) Suppose that only one sphere (E) swings to the right with speed v_E just after collision. By finding v_E , hence explain why it is not possible for only one sphere to swing to the right. Given that the speed of sphere A and B just before collision is 0.50 m s^{-1} . The mass of each sphere is 0.020 kg . (3 marks)
- (b)(ii) The collision repeats alternately from either side for a period of time before the system eventually returns to rest. Discuss whether the collisions are perfectly elastic. (2 marks)

Official marking scheme

Solution	Marks	Remarks
10. (a) 1. Use the protractor to measure or mark a specific angle at which the ball is released. Make sure the angle is measurable and recorded.	1A	
2. Hold the ball at the marked angle and release it from rest with the string taut, allowing it to swing like a pendulum.	1A	
3. Measure the angle at which the ball reaches its highest point and record the value.	1A	
Or Observe the angle at which the ball swings back to the original releasing position.		
4. Compare the measurements of the ball's initial angle of release to the angle it reaches on the opposite side of the swing. (The angles equal implies total mechanical energy is conserved.)	1A	
	4	
(b) (i) (I) The tension forces are vertical / perpendicular to the direction of collision, the condition of no external forces along the colliding / horizontal direction is still fulfilled, the law can be applied.	1A	
Or The tensions in the string acting on the spheres are balanced by the weights of the spheres so that the net force acting on the system is zero.		
	1	
(II) By conservation of total momentum, v_E is $2 \times 0.50 = 1.0 \text{ m s}^{-1}$	1A	
However, the final total kinetic energy would then be		
$\frac{1}{2}(0.020)(1.0)^2 = 0.01 \text{ J} > \frac{1}{2}(2 \times 0.020)(0.50)^2 = 5 \times 10^{-3} = 0.005 \text{ J}$	1M	
which is greater than the total kinetic energy before collision, thus it is not possible.	1A	
	3	
(ii) Not perfectly elastic as some kinetic energy is lost as sound / thermal energy, the total kinetic energy is not conserved.	1A+1A	
	2	

(a) Measure and record the initial angle from the vertical with the protractor. Hold the ball at this angle with its string taut and release it from rest without pushing. Measure the maximum angle reached on the opposite side. Compare the two angles: approximately equal angles imply approximately equal maximum heights and gravitational potential energies, consistent with conservation of mechanical energy when losses are small.

(b)(i)(I) During the short collision, the strings are vertical and their tensions have no horizontal component. The external horizontal impulse is negligible, so total horizontal momentum is conserved.

(b)(i)(II) If only E moved, momentum conservation would give $2m(0.50) = mv_E$, so $v_E = 1.0 \text{ m s}^{-1}$. Initially $K = \frac{1}{2}(2 \times 0.020)(0.50)^2 = 0.005 \text{ J}$; the proposed final $K = \frac{1}{2}(0.020)(1.0)^2 = 0.010 \text{ J}$. This would create kinetic energy without an energy source, so that outcome is impossible.

(b)(ii) The collisions are not perfectly elastic. Some kinetic energy becomes sound and thermal energy, so the total kinetic energy after each collision is smaller. With air resistance neglected, this explains the gradual reduction of the motion.

Question 11

Read the following passage about deformation sensors and answer the questions that follow.

Figure 11.1(a) shows a simple sensing device consisting of a thin, flat rectangular metal foil attached to a plastic backing plate. When this foil, length l and width w , is stretched to deform as shown in Figure 11.1(b), its resistance will change.

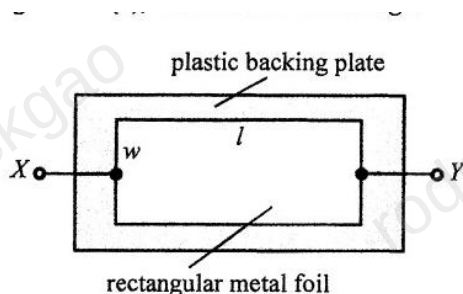


Figure 11.1(a)

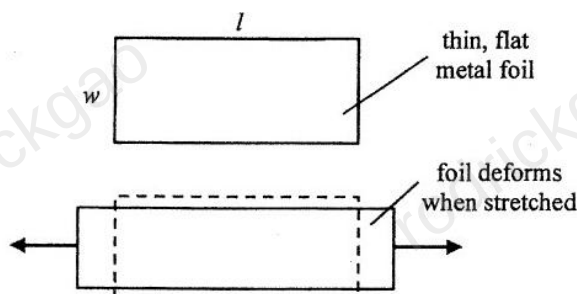


Figure 11.1(b)

Figure 11.1

When this device is connected to a suitable circuit, it can serve as a sensor to monitor deformation which results in a change in parameters of the circuit (such as voltage).

- (a) Referring to Figure 11.1(a) and (b), state and explain how the metal foil's resistance across XY changes when the foil is stretched. Assume that the thickness and resistivity of the foil remain unchanged. (2 marks)
- (b) Figure 11.2 shows a circuit with four resistors of resistance R_1 , R_2 , R_3 and R_4 connected to an input voltage V_{in} . V_{out} is the voltage across DB.

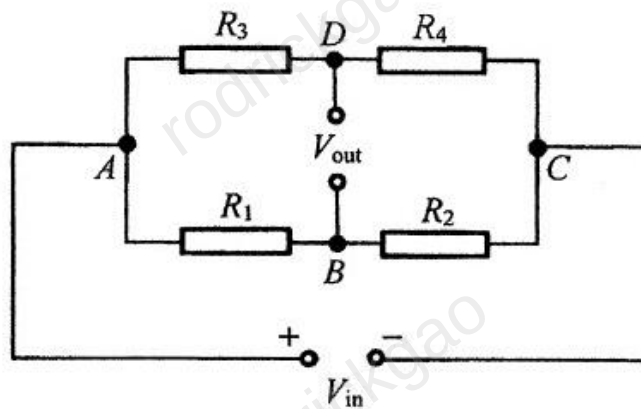


Figure 11.2

- (i) Current I_1 flows through branch ABC. Write an equation relating V_{in} , I_1 , R_1 and R_2 . (1 mark)
- (ii) R_4 in the circuit is the sensing device described in the passage, with unstretched resistance $500\ \Omega$. Find the corresponding percentage change in V_{out}/V_{in} when R_4 increases 0.2% (i.e. becomes $501\ \Omega$). (2 marks)

Given: $V_{out}/V_{in} = R_4/(R_3 + R_4) - R_2/(R_1 + R_2)$; $R_1 = R_2 = R_3 = 470\ \Omega$.

- (iii) Sensing devices S_1 , S_2 , S_3 and S_4 connected to a suitable circuit are attached to ceiling ABCD as shown in Figure 11.3 in order to monitor the growth of cracks ①, ② and ③. After a few weeks, the extent of deformation detected is in the order $S_1 > S_2 > S_3 > S_4$.

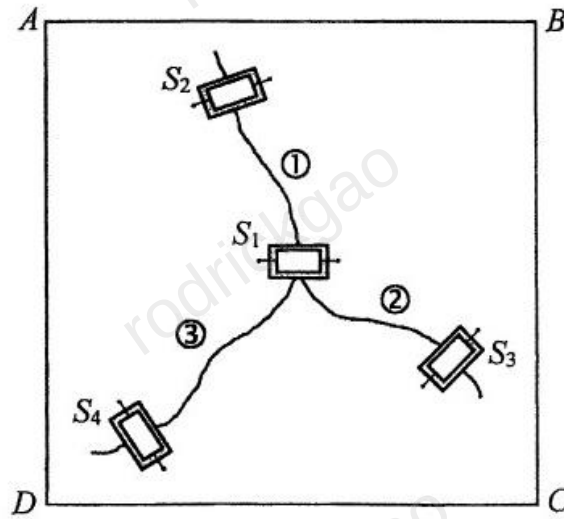


Figure 11.3

- (b)(iii)(I) Deduce which crack (①, ② or ③) has grown the most. (1 mark)
- (b)(iii)(II) Deduce the most probable stretching direction (AB, BC or AC) along the ceiling. (1 mark)

Official marking scheme

Solution	Marks	Remarks
11. (a) As the foil becomes longer / l increases while its cross-sectional area becomes smaller / A decreases due to decrease in width w , its resistance would increase according to $R = \frac{\rho l}{A}$.	1A 1A 2	
(b) (i) $V_{\text{in}} = I_1(R_1 + R_2)$	1A 1	
(ii) Before: $\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{500}{470 + 500} - \frac{470}{470 + 470} = 0.0154639$ After: $\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{501}{470 + 501} - \frac{470}{470 + 470} = 0.0159629$ Thus $\frac{V_{\text{out}}}{V_{\text{in}}}$ increases $3.22687\% \approx 3.23\%$.	1M 1A 2	Accept $\pm(3.2 \sim 3.23)\%$ as the change
(iii) (I) crack ① (II) along AB	1A 1A 2	

(a) $R = \rho l/A$. Stretching increases l and decreases the width, hence decreases cross-sectional area A because the thickness is unchanged. With ρ unchanged, both changes increase the resistance across XY .

(b)(i) The resistors R_1 and R_2 carry the same branch current I_1 , so $V_{\text{in}} = I_1(R_1 + R_2)$.

(b)(ii) Initially $V_{\text{out}}/V_{\text{in}} = 500/970 - 470/940 = 0.0154639$. After stretching it is $501/971 - 470/940 = 0.0159629$. Percentage increase = $(0.0159629 - 0.0154639)/0.0154639 \times 100\% = 3.23\%$. This is a relative percentage change, not the percentage-point difference or the 0.2% change in R_4 .

(b)(iii)(I) Crack ① has grown the most: the greatest deformation is detected by S_1 across that crack.

(b)(iii)(II) The most probable stretching direction is along AB , consistent with the relative deformations of the differently oriented sensors.

Question 12

Figure 12.1 shows a coil of several turns.



Figure 12.1

- (a) Sketch the magnetic field pattern produced by the coil when it carries a current I as shown. (2 marks)
- (b) As shown in Figure 12.2(a), this coil is now connected to an a.c. source to serve as a transmitter coil (T). A receiver coil (R) connected to a CRO is placed directly above T . The time variation of the voltage of coil R is displayed on the CRO (Figure 12.2(b)).

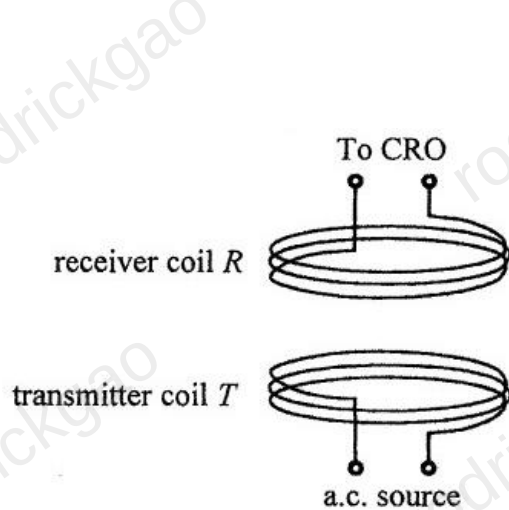


Figure 12.2(a)

CRO display

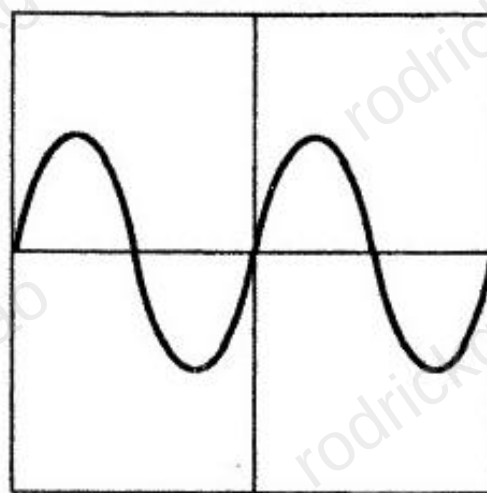
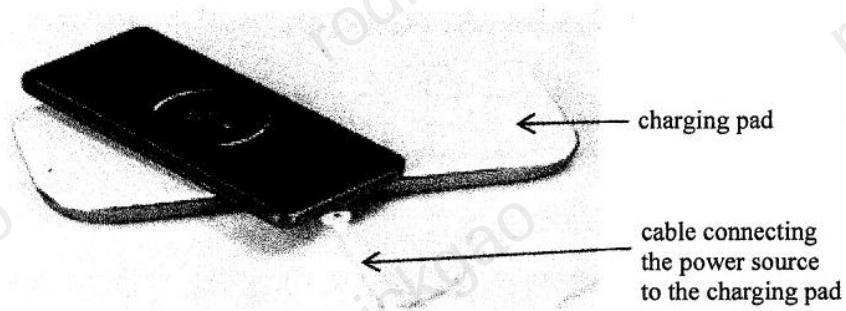


Figure 12.2(b)

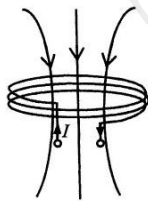
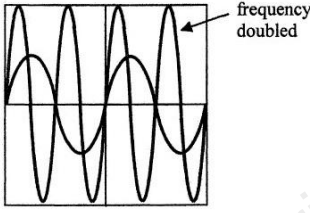
- (i) Explain how the voltage of the receiver coil R is produced. (2 marks)
- *(ii) On Figure 12.2(b), sketch the CRO display when the frequency of the a.c. source is doubled. (2 marks)

- (c) A mobile phone with a receiver coil inside can be charged wirelessly by placing it on a charging pad equipped with a built-in transmitter coil, which is connected to a power source.



Explain why wirelessly charging of phones need phone cases with non-metallic back covers. (2 marks)

Official marking scheme

Solution	Marks	Remarks
12. (a)		
	2A	1A for correct direction (also accept field lines pointing upwards for observers with different perception of current direction) 1A for correct field pattern
	2	
(b) (i) The a.c. flowing in transmitter coil <i>T</i> generates a changing magnetic field, thus by electromagnetic induction an induced e.m.f. is produced in receiver coil <i>R</i> to oppose the changing magnetic flux experienced by it.	1A 1A	
	2	
(ii)		
CRO display 	2A	1A for correct frequency 1A for correct amplitude
	2	
(c) If the back cover is metallic, eddy currents will be produced (by induction), loss of energy / heating effect results / magnetic field / flux is blocked by the metal case and thus making wireless charging impossible. Or Non-metallic materials such as glass are insulators, and no eddy currents are produced. As a result, energy loss / heating effect will be minimized / the magnetic field can pass through the back cover easily.	1A 1A	
	2	

(a) Draw field lines passing through the centre approximately parallel to the coil axis and curving around outside into closed loops. Apply the right-hand grip rule to the indicated current; in the marking diagram the arrows through the centre point downwards. The direction must be consistent throughout the pattern.

(b)(i) Alternating current in *T* produces a changing magnetic field and changing magnetic flux through *R*. By Faraday's law, $\varepsilon = -N \Delta\Phi/\Delta t$, this induces an alternating voltage in *R*; its polarity opposes the flux change.

(b)(ii) Draw a sinusoidal trace with twice the frequency, half the period and twice the original amplitude, as in the marking scheme. For the same flux amplitude, the maximum rate of flux change, and therefore the induced-voltage amplitude, is proportional to frequency.

(c) A metallic back cover would develop eddy currents in the changing magnetic field. These dissipate energy as heat and oppose the field passing through to the receiver coil, reducing charging efficiency. A non-metallic insulating back cover avoids these eddy-current losses.

Question 13

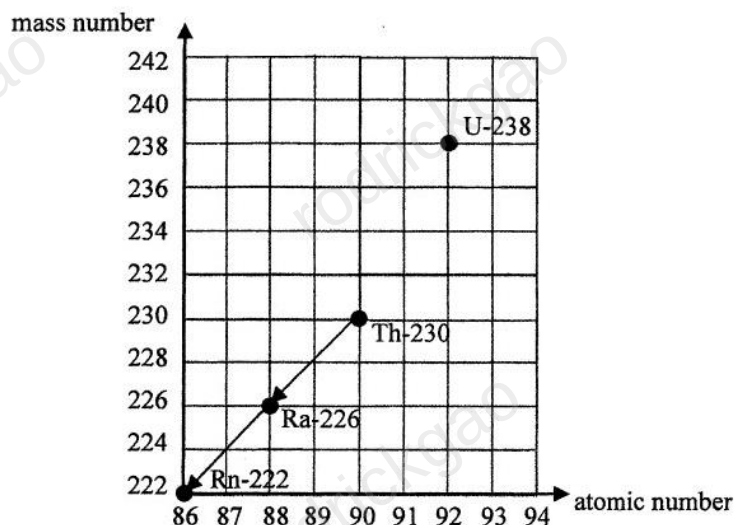
Radon (Rn-222) is a radioactive gas with no colour, no smell and no taste. It is released from bedrock material underground or from concrete composed of granite. Rn-222 decays with a half-life of 3.82 days by giving out α particles.

(a) * (i) Find the decay constant (unit: s^{-1}) of Rn-222. (2 marks)

(ii) Inside buildings, the average activity due to radon in 1 m^3 of air is about 48 Bq. Estimate the average number of radon atoms in 1 m^3 of air inside buildings. (2 marks)

(iii) Why is indoor radon considered harmful to human although α particles have limited penetrating power? (2 marks)

(b) Rn-222 belongs to a decay series starting from uranium-238 (U-238). It is known that U-238 undergoes decay in the sequence α - β - β - α to become thorium-230 (Th-230). Complete the decay series below from U-238 to Th-230 using arrows. (2 marks)



Official marking scheme

Solution	Marks	Remarks
13. (a) (i) $k = \frac{\ln 2}{t_{1/2}}$ $= \frac{\ln 2}{3.82 \times 24 \times 3600}$ $= 2.1001405 \times 10^{-6} \text{ s}^{-1} \approx 2.10 \times 10^{-6} \text{ (s}^{-1}\text{)}$	1M 1A 2	
(ii) $A = kN$ $N = \frac{48}{2.10 \times 10^{-6}}$ $= 2.285561 \times 10^7 \approx 2.29 \times 10^7$	1M 1A 2	e.c.f. from (a)(i) Accept: $(2.28 \sim 2.3) \times 10^7$
(iii) As radon is a gas that can be inhaled into the lungs of human, the relatively strong ionizing power of α particles emitted by decaying radon / radioactive gas might affect the organs / cells nearby.	1A 1A 2	
(b) 	2A 2	

(a)(i) Convert the half-life to seconds: $T = 3.82 \times 24 \times 3600 = 330048 \text{ s}$. The decay constant is $k = \ln 2/T = 2.10 \times 10^{-6} \text{ s}^{-1}$.

(a)(ii) Activity $A = kN$, so $N = 48/(2.10014 \times 10^{-6}) = 2.29 \times 10^7$ atoms per cubic metre. Bq means decays per second, not the number of atoms.

(a)(iii) Radon gas can be inhaled into the lungs. Alpha particles emitted inside the body have strong ionizing power and deposit energy in nearby living cells; their short range does not make an internal source harmless.

(b) Alpha decay reduces mass number by 4 and atomic number by 2; beta-minus decay leaves mass number unchanged and increases atomic number by 1. The sequence is ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} \rightarrow {}^{234}_{91}\text{Pa} \rightarrow {}^{234}_{92}\text{U} \rightarrow {}^{230}_{90}\text{Th}$, with emissions $\alpha, \beta^-, \beta^-, \alpha$ respectively. On the (atomic number, mass number) graph, join $(92, 238) \rightarrow (90, 234) \rightarrow (91, 234) \rightarrow (92, 234) \rightarrow (90, 230)$. Use these numerical coordinates: some plotted labels in the supplied marking scan are vertically misaligned with the mass-number grid.